

The triangular H^2 conjecture (TLMC 00000001147) is false:
 $\dim H^2(\mathfrak{aff}(1)) = 0$

1 The conjecture

Quoted verbatim from the problem statement:

Definition: A triangular Lie algebra is a subalgebra of triangular matrices. Conjecture: The dimension of the second cohomology of a triangular Lie algebra is two; and the dimension is realized by one contribution each from the outer derivations and the center. (triangular H^2 dimension two)

Cohomology means Chevalley–Eilenberg cohomology $H^\bullet(\mathfrak{g}) = \text{Ext}_{U(\mathfrak{g})}(\mathbb{k}, \cdot)$, computed by the standard complex

$$C^q = \text{Hom}(\Lambda^q \mathfrak{g}, \mathbb{k}), \quad (\delta^q \omega)(x_1, \dots, x_{q+1}) = - \sum_{i < j} (-1)^{i+j} \omega([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{q+1}).$$

2 The counterexample

Take the two-dimensional non-abelian Lie algebra

$$\mathfrak{aff}(1) = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{k} \right\}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad [H, E] = E.$$

It is a subalgebra of the algebra of upper triangular 2×2 matrices, hence a *triangular Lie algebra* under the definition quoted above. We claim

$$\dim H^2(\mathfrak{aff}(1)) = 0 \neq 2.$$

2.1 The cochain complex

The cochain spaces are

$$C^1 = \text{Hom}(\mathfrak{g}, \mathbb{k}), \quad \dim C^1 = 2; \quad C^2 = \text{Hom}(\Lambda^2 \mathfrak{g}, \mathbb{k}), \quad \dim C^2 = 1; \quad C^3 = \text{Hom}(\Lambda^3 \mathfrak{g}, \mathbb{k}) = 0,$$

because $\dim \mathfrak{g} = 2$ forces $\Lambda^3 \mathfrak{g} = 0$. An alternating form is determined by its value on the single wedge basis vector $H \wedge E$; we identify $C^2 \cong \mathbb{k}$ via $\omega \mapsto \omega(H, E)$.

2.2 The differential is onto

For $f \in C^1$,

$$(\delta^1 f)(H, E) = -f([H, E]) = -f(E).$$

The map $f \mapsto \delta^1 f$ is therefore nonzero ($f = E^*$ gives $(\delta^1 f)(H, E) = -1$), and since $\dim C^2 = 1$ it is surjective. Hence

$$B^2 = \text{im } \delta^1 = C^2.$$

2.3 Every 2-cochain is a cocycle

Since $C^3 = 0$, we have $\delta^2 = 0$ and

$$Z^2 = \ker \delta^2 = C^2.$$

2.4 Conclusion

Claim 1. $\dim H^2(\mathfrak{aff}(1)) = \dim Z^2 - \dim B^2 = 1 - 1 = 0 \neq 2$. *The conjecture is false.*

The secondary clause fails as well: $\text{Der}(\mathfrak{aff}(1)) = \{D : D(H) = bE, D(E) = cE\}$ has dimension 2, $\text{Inn} = [\mathfrak{g}, \mathfrak{g}]$ -conjugates give $\dim \text{Inn} = 1$, so $\dim \text{Out} = 1$, while the center $Z(\mathfrak{aff}(1)) = 0$ has dimension 0; there is no “one plus one” realization of a zero-dimensional space.

3 The conjecture is not vacuous either

The Heisenberg algebra $\mathfrak{n}_3 = \langle x, y, z \mid [x, y] = z \rangle$ (strictly upper triangular 3×3 matrices, also triangular under the given definition) satisfies $\dim C^2 = 3$, $\delta^2 \equiv 0$ (all brackets land in the center, so $(\delta^2 \omega)(x, y, z) = -\omega(z, z) = 0$), $\dim B^2 = 1$, whence $\dim H^2(\mathfrak{n}_3) = 2$: the conjecture’s value is attained by some triangular Lie algebras but is not an identity across the class.

4 Machine verification

- `reproduce.py` recomputes $\dim Z^2, B^2, H^2$ from the structure constants with exact rational Gaussian elimination: it prints $\dim H^2(\mathfrak{aff}(1)) = 0$ and $\dim H^2(\mathfrak{n}_3) = 2$.
- `lean4/Main.lean` formalizes the computation in bare Lean 4 (no Mathlib, coefficients in $\mathbb{Z}$; ranks are field-independent): surjectivity of the differential (`d1_surj`), vanishing of degree 3 (`no_strict_inc`: no strictly increasing triple of values in a two-element index set, i.e. $\Lambda^3 \mathfrak{g} = 0$), triviality of the cohomology (a single class, `cobdy_all`), and the key dimension obstruction (`no_four_classes`): a 2-dimensional space over any field contains the four distinct points $0, v, w, v + w$ for a basis (v, w) , so $\dim H^2 = 2$ would exhibit four pairwise non-cohomologous 2-cochains — impossible when all pairs are cohomologous.
- `lean4/Check.lean` audits every theorem with `#print axioms`: all of `Br_HE`, `d1_surj`, `d1_add`, `no_strict_inc`, `cobdy_all`, `no_four_classes`, `main` **do not depend on any axioms** (no `sorryAx`, no `Classical.choice`, no `propext`, no `Quot.sound`).